Step-by-Step Development Plan for ArchAI on Unity 6

Phase 1: Setup & Core Architecture

✅ **Set up Unity 6 Project & Code Structure**

- Initialize a new **Unity 6 project** (HDRP for realistic rendering, URP for performance).
- Organize core scripts:
  - **Game Manager** (Handles object placement, scene transitions, input handling).
  - **UI Manager** (Handles user input from the home design survey).
  - **Data Manager** (Stores and retrieves user preferences).

✅ **Survey Input & Data Handling**

- Implement a UI system for **Google Forms-style survey**.
- Store user responses in **ScriptableObjects** or **JSON** for easy access.

✅ **2D Grid System for Architectural Layouts**

- Implement a **grid-based system** for floor plans (like a tilemap).
- Set up **rule-based constraints** for **room placement** (e.g., kitchens next to dining rooms).

Phase 2: AI-Driven 2D Floor Plan Generation

✅ **Rule-Based Shape Grammar System**

- Implement **shape grammars** to generate room layouts dynamically.
- Encode **architectural constraints** (entryways, window placements, etc.).

✅ **Graph-Based Layout Optimization**

- Implement **graph algorithms** to optimize room adjacency and layout efficiency.
- Use *A Pathfinding** to ensure **smooth navigation** inside the house.

✅ **AI-Driven Layout Generation**

- Train a **Neural Network (or Reinforcement Learning)** model for space planning.
- Optimize room placement based on **real-world architectural layouts**.

Phase 3: Building Code Compliance Checker

✅ **Automated Rule-Based Compliance System**

- Implement a **rule validation system** to check:
  - Minimum bedroom sizes.
  - Proper ventilation and exit placements.
  - Fire safety compliance.
- Display **warnings & alternative suggestions** for invalid designs.

Phase 4: 3D Model Generation & Rendering

✅ **Procedural 3D House Generation**

- Convert **2D layouts** → **3D modular prefabs**.
- Implement **procedural generation** for walls, doors, and furniture.

✅ **Optimized 3D Rendering**

- Use **GPU-accelerated rendering** for **real-time previews**.
- Implement **LOD (Level of Detail) optimization** for large buildings.

✅ **Material & Texture Automation**

- Assign materials based on user-selected **styles & preferences**.
- Implement **PBR (Physically-Based Rendering)** for realistic visuals.

Phase 5: UI/UX & Final Optimizations

✅ Drag-and-Drop Customization

- Users can **modify room placements** and **resize spaces** interactively.
- Implement **undo/redo functionality**.

✅ Performance & Bug Testing

- Test on different devices (PC, Web, Mobile).
- Optimize rendering and processing speeds.

✅ Final Deployment

- Package ArchAI for **Web & Desktop**.